

Cbits
 $|00\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$ Tensor/outer product: $V \otimes W = VW^T$

You start by saying you have a binary string of length k

So you have 2^k possible things. each $|0 \dots 1 \dots\rangle$ is a basis in a space $\mathbb{R}^{2^k}$

- Reversible means for $Ax=b$ given A, b , you can uniquely find x
- Quantum Computers only use reversible operations

CNOT: On Pairs of bits, one is "control" bit, other is target bit

$$\begin{array}{cc} 00 \rightarrow 00 \\ 01 \rightarrow 01 \\ 10 \rightarrow 11 \\ 11 \rightarrow 10 \end{array} \quad C = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$|\psi\rangle$: Vector
 $\langle\psi|$: Hermitian of vector
 $\langle\psi|\phi\rangle$: Inner product
 $|\psi\rangle\langle\phi|$: Outer product

Qbits are represented by $\begin{pmatrix} a \\ b \end{pmatrix}$ $a, b \in \mathbb{C}$ $\|a\|^2 + \|b\|^2 = 1$

• Cbit special example

• Superposition: When we measure qbit (end of computation), collapses to either 0 or 1
 • Collapse to 0 with prob $\|a\|^2$ & 1 with prob $\|b\|^2$

• Use Matrices to operate on Qbits:

Hadamard gate: $\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$ $H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$ $H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$
 Grover's Alg: $H^{\otimes n} = H \otimes H \dots H$ n times
 $|0^n\rangle = |0\rangle^{\otimes n} = |0\rangle \otimes |0\rangle \dots$

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Say $|w\rangle$ is the correct state
 Say $|w^\perp\rangle$ are superpos of incorrect states

$|w^\perp\rangle = \frac{1}{\sqrt{N-1}} \sum_{x \neq w} |x\rangle$ $|s\rangle = \sin(\theta)|w\rangle + \cos(\theta)|w^\perp\rangle$
 $\sin \theta = \frac{1}{\sqrt{N}}$

2. $U_w|x\rangle$ entire job is to recognize $|w\rangle$

$U_w|x\rangle = \begin{cases} -|x\rangle & x=w \\ |x\rangle & x \neq w \end{cases}$

$U_w = I - 2|w\rangle\langle w|$ reflect across $|w^\perp\rangle$

3. Diffusion Operator: $|s\rangle\langle s| - I$ Inversion about the mean

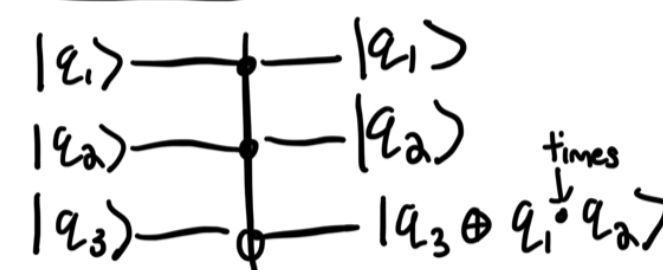
$\alpha_w \rightarrow 2\mu - \alpha_w$ (u get that 3θ change to w axis from initial θ)
 $\alpha_w \rightarrow 2\mu - \alpha_w$
 $\psi: (\vec{x}, t): \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{C}$ like $\frac{d\vec{x}}{dt} = A\vec{x}$

Time Dependent Schrodinger: $i\hbar \frac{\partial}{\partial t} \psi(\vec{x}, t) = \hat{H} \psi(\vec{x}, t)$ like $\frac{d\vec{x}}{dt} = A\vec{x}$

$\hat{H} = \frac{\hbar^2}{2m} \nabla^2 + V(x)$ (one electron)
 $E = \text{total Energy} = \frac{p^2}{2m} + V$

2 particles, either fermions or bosons
 • Fermions can only occupy one state at a time
 • State can be anything, like momentum, energy, spin etc.
 • Bosons can have multiple of the same state

Toffoli Gate (Controlled CNOT)



Rotation Gates

Rotation of θ around
 X-axis: $R_x(\theta) = \begin{pmatrix} \cos(\frac{\theta}{2}) & -i\sin(\frac{\theta}{2}) \\ i\sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{pmatrix}$
 Y-axis: $R_y(\theta) = \begin{pmatrix} \cos(\frac{\theta}{2}) & -\sin(\frac{\theta}{2}) \\ \sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{pmatrix}$
 Z-axis: $R_z(\theta) = \begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{pmatrix}$

Spin up - $|0\rangle$
 Spin down - $|1\rangle$

Quantum Gates

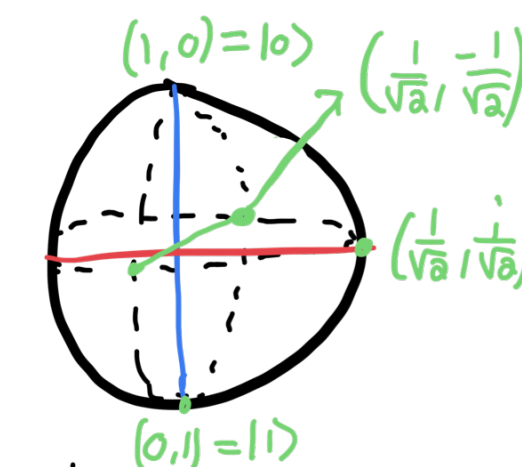
$|1\rangle \xrightarrow{\text{Pauli X gate}} |0\rangle$ $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ 4^{deg} $|\psi\rangle = a|0\rangle + b|1\rangle$
 2^{deg} $|\psi\rangle = \cos(\frac{\theta}{2})|0\rangle + e^{i\phi} \sin(\frac{\theta}{2})|1\rangle$

Hadamard: $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

- These logic gates must be
- Hermitian: $U = U^\dagger$
- Unitary: $UU^\dagger = I$

$|0\rangle \xrightarrow{Y} |1\rangle$ $Y|0\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ i \end{bmatrix} \equiv i|1\rangle$
 $|1\rangle \xrightarrow{Z} |-1\rangle$ $Z|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix} \equiv |-1\rangle$

Bloch Sphere



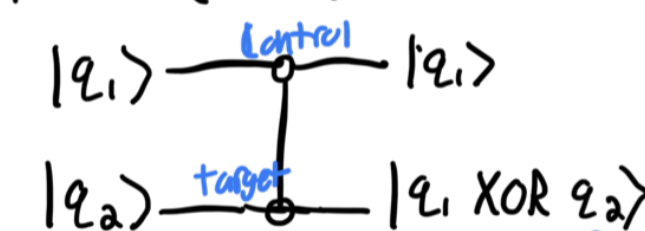
Spin $\frac{1}{2}$ rotation operator: $D(\hat{n}, \phi) = \cos(\frac{\phi}{2})I - i\sin(\frac{\phi}{2})\vec{\sigma} \cdot \hat{n}$

For a general n -qubit system

$|\psi\rangle = \sum_{k=1}^{2^n} c_k |\phi_k\rangle$: Superposition

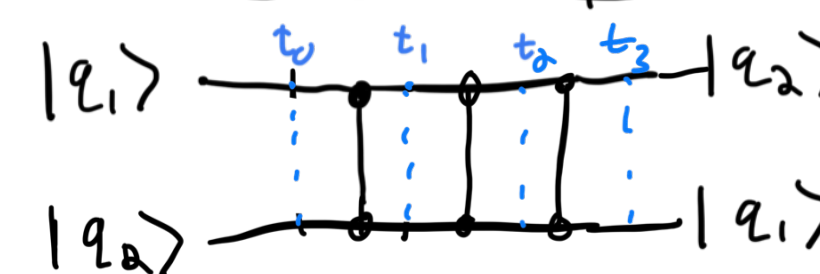
$U = \sum_{j,k=1}^{2^n} U_{jk} |\phi_j\rangle\langle\phi_k|$: Gate operator

CNOT (XOR) (2 Qubits)



$U_{\text{CNOT}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

2 Qubit Swap



$t_0: |q_1, q_2\rangle$
 $t_1: |q_1, q_1 \text{ XOR } q_2\rangle$
 $t_2: |q_1 \oplus (q_1 \oplus q_2), q_1 \oplus q_2\rangle$
 $t_3: |q_2, q_2 \oplus (q_1 \oplus q_2)\rangle = |q_2, q_1\rangle$

Wave eq: $A \cos(Kx - \omega t)$

$\cos(x)$ vs $\cos(\omega x)$ (tighter waves)

X-wt. Wave peak travels velocity t from x .

$e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots$
 $(e^{-i\theta\hat{H}})^* (e^{-i\theta\hat{H}}) = I$ $\hat{H}$ is a Hermitian matrix
 $U = e^{-i\theta\hat{H}}$

"Small change $\hat{H}$ propagated by e , scaled by $\theta = U$ "